WORKED SOLUTION

STEP II 2016, Question 8

Pure Mathematics • Approximating sums using integrals

STEP II 2016, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q8

The key idea

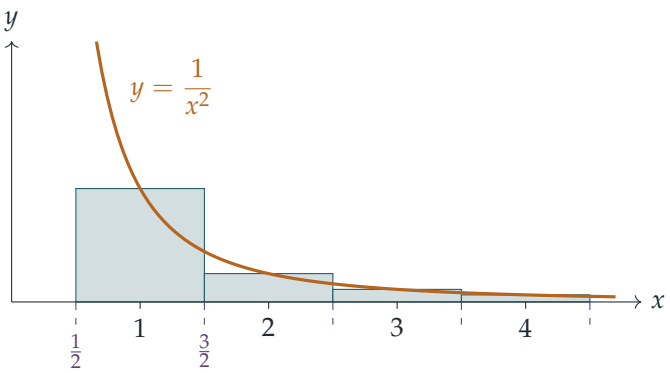
The term $\frac{1}{r^2}$ is the area of a rectangle of width 1 and height $\frac{1}{r^2}$, and if that rectangle is centred on $x = r$ its area is very close to the area under $y = \frac{1}{x^2}$ between $r - \frac{1}{2}$ and $r + \frac{1}{2}$ (the midpoint rule). Gluing these strips together turns a sum into one integral. Then the error in each strip can be computed exactly, and adding up the errors is what produces a second series, $\sum \frac{1}{r^4}$, almost for free.

Solution

The integral and the sketch

For $m > \frac{1}{2}$ the lower limit is positive, so the integrand is continuous on the range and

$$\int_{m-\frac{1}{2}}^{\infty} \frac{1}{x^2}\,\mathrm{d}x = \lim_{X\to\infty}\left[-\frac{1}{x}\right]_{m-\frac{1}{2}}^{X} = \frac{1}{m-\frac{1}{2}} = \frac{2}{2m-1}.$$



The rectangle centred on $x = r$ has width 1 and height $\frac{1}{r^2}$, so its area is $\frac{1}{r^2}$. Over the strip $r - \frac{1}{2} \le x \le r + \frac{1}{2}$ the curve lies above the rectangle's top on the left half and

below it on the right half, and these two small pieces nearly cancel. So

$$\frac{1}{r^2} \approx \int_{r-\frac{1}{2}}^{r+\frac{1}{2}} \frac{1}{x^2}\,\mathrm{d}x.$$

Adding these for $r = m, m+1, \ldots, n$, the strips fit together end to end to cover $[m - \frac{1}{2}, n + \frac{1}{2}]$, which gives $(*)$.

Part (i)

Let $n \to \infty$ in $(*)$. The right-hand side tends to the integral we evaluated, so

$$\sum_{r=m}^{\infty} \frac{1}{r^2} \approx \frac{2}{2m-1}.$$

We can apply this to the whole series, or add the first few terms exactly and approximate only the tail:

$$\begin{aligned}
m = 1: &\quad E \approx \frac{2}{1} = 2,\\
m = 2: &\quad E = 1 + \sum_{r=2}^{\infty} \frac{1}{r^2} \approx 1 + \frac{2}{3} = \frac{5}{3},\\
m = 3: &\quad E = 1 + \frac{1}{4} + \sum_{r=3}^{\infty} \frac{1}{r^2} \approx \frac{5}{4} + \frac{2}{5} = \frac{33}{20}.
\end{aligned}$$

These are the three required approximations. They should improve as m increases, since the curve is flatter further out and the midpoint approximation is better there.

Part (ii)

The integral over one strip is

$$\int_{r-\frac{1}{2}}^{r+\frac{1}{2}} \frac{1}{x^2}\,\mathrm{d}x = \frac{1}{r-\frac{1}{2}} - \frac{1}{r+\frac{1}{2}} = \frac{1}{r^2-\frac{1}{4}} = \frac{4}{4r^2-1}.$$

So the error in approximating $\frac{1}{r^2}$ by this integral is

$$\frac{4}{4r^2-1} - \frac{1}{r^2} = \frac{4r^2-(4r^2-1)}{r^2(4r^2-1)} = \frac{1}{r^2(4r^2-1)} = \frac{1}{4r^4} \cdot \frac{1}{1-\frac{1}{4r^2}}.$$

When r is large, $\frac{1}{4r^2}$ is small, so the last factor is close to 1 and the error is approximately $\frac{1}{4r^4}$, as required. (The integral is always the larger, which fits the picture: $\frac{1}{x^2}$ is convex, so the extra area on the left of each strip beats the shortfall on the right.)

For the final result, add up the errors from $r = m$ onwards. The strip integrals add up to the integral we evaluated first, so

$$\frac{2}{2m-1} - \sum_{r=m}^{\infty} \frac{1}{r^2} = \sum_{r=m}^{\infty} \frac{1}{r^2(4r^2-1)} \approx \frac{1}{4} \sum_{r=m}^{\infty} \frac{1}{r^4}.$$

Ask yourself Why not take $m = 1$, which looks simplest?

With $m = 1$ the left-hand side is $2 - 1.645 = 0.355$, giving $\sum \frac{1}{r^4} \approx 1.42$, which is far too big. The trouble is that "$\approx \frac{1}{4r^4}$" is only good for large r: at $r = 1$ the true error is $\frac{1}{3}$, not $\frac{1}{4}$, and that first term dominates the whole sum. So we should keep the first few terms exact and use the approximation only on the tail. Taking $m = 3$ (the choice that gave $\frac{33}{20}$), the worst factor $\frac{1}{1-1/(4r^2)}$ is $\frac{36}{35}$, within 3% of 1:

$$\frac{1}{4}\sum_{r=3}^{\infty}\frac{1}{r^4} \approx \frac{2}{5} - \left(E - 1 - \frac{1}{4}\right) \approx 0.4 - 0.395 = 0.005,$$

so $\sum_{r=3}^{\infty}\frac{1}{r^4} \approx 0.02$ and

$$\sum_{r=1}^{\infty}\frac{1}{r^4} \approx 1 + \frac{1}{16} + 0.02 = 1.0825 \approx 1.08.$$

This is the required result. Since E is only given to three decimal places, an error of up to 0.0005 in E becomes up to 0.002 here, which still leaves 1.08.

Checks

- The approximations 2, $\frac{5}{3} \approx 1.667$, $\frac{33}{20} = 1.65$ decrease towards $E = \frac{\pi^2}{6} \approx 1.6449$, and each is too big, as the convexity argument predicts.
- At $r = 10$ the exact error $\frac{1}{100\times399} \approx 2.506 \times 10^{-5}$ is very close to $\frac{1}{4\times10^4} = 2.5 \times 10^{-5}$.
- The exact value is $\sum \frac{1}{r^4} = \frac{\pi^4}{90} \approx 1.0823$, which agrees with our 1.0825. Using $m = 2$ instead gives $1 + 4\left(\frac{2}{3} - 0.645\right) \approx 1.087$, noticeably worse; $m = 1$ gives 1.42.

Common slips

- Using $m = 1$ in the last part, where the large-r approximation $\frac{1}{4r^4}$ is poor, and getting 1.42.
- Getting the sign of the error backwards: the integral exceeds $\frac{1}{r^2}$, so $\frac{2}{2m-1}$ is an overestimate of the tail.
- Forgetting to add the exact first terms ($1 + \frac{1}{4}$, and $1 + \frac{1}{16}$) back on.

Beyond STEP

This is the midpoint rule, and the error term $\frac{1}{4r^4}$ is a glimpse of the Euler–Maclaurin formula, which corrects a sum by an integral plus a series of derivative terms; it is how sums like $\sum \frac{1}{r^2}$ are computed to many decimal places. The exact values $\frac{\pi^2}{6}$ and $\frac{\pi^4}{90}$ come

from Fourier series or from Euler's product for $\sin x$. Abbott's *Understanding Analysis* treats series and integrals carefully, and Siklos's *Advanced Problems in Mathematics* has more questions in this spirit.

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